

Statistical Data Analysis

Lecture 1

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Where Are We?

📍 Chapter 1: Introduction

📍 Chapter 2: Summarizing data

- Section 2.1 – Data
- Section 2.2 – Summarizing data

▶▶ Chapter 3: Exploring distributions

Introduction

Motivation

- › What is **statistics**?
 - › Relies on **math**
 - › Concerns collecting, visualising, summarizing, analysing, modeling, interpreting, ... **data**
 - › **Descriptive** vs. **inferential**
- › It has applications in a variety of fields
 - › Medical studies, politics, climate change research, finance, and many more!
- › It is a core part of **machine learning** and **data science**
 - › See [**this**](#) tweet (are they still called tweets? Anyway, the link goes to a Reddit thread) for a joke by Arvind Narayanan from 2018

› Stages

- › Research question formulation
- › Experimental design
- › Data collection
- › Data analysis
- › Results interpretation
- › Results presentation

› Necessary skills for a statistician

- › Being knowledgeable about statistical **theory** (this what lectures and Syllabus are for)
- › Being able to translate a **practical** problem in statistical terms, perform an appropriate statistical analysis, interpret and effectively present its results (this is what  and the assignments are for)

Data & Variables

Data & Variables

- › **Data** are the quantified results of a study
 - › A collection of data generally consists of characteristics of observation units » **variable realizations**
- › There are different types of data:

Name	Sex	O.W.L. Potions Grade	Year of Birth	Nr. of pets	Height (in cm)
Harry	M	Exceeds Expectations	1980	1	176.5
Ron	M	Exceeds Expectations	1980	1	187.6
Hermione	F	Outstanding	1979	1	165.1

- › Variables can differ in several ways
 - › Measurement **scales**
 - › Discrete vs. continuous domain
 - › Univariate vs. multivariate

Measurement Scales - Qualitative

> Nominal (e.g. **Name**)

- > It can only be decided if values are equal or not
- > There is no intrinsic ordering of the categories
- > Appropriate operations: comparison

> Ordinal (e.g. **O.W.L. Potions Grade**)

- > Values can be compared
- > There is an intrinsic ordering of the categories
- > But the distance between categories has no interpretable meaning
- > Appropriate operations: comparison and ordering

Measurement Scales - Quantitative

› Interval (e.g. **Year of Birth**)

- › The distance between values has an interpretable meaning, but the ratio has not
- › There is no **true** zero point
- › Appropriate operations: comparison, ordering, **+** and **-**

› Ratio (e.g. **Height (in cm)**)

- › The ratio between values has an interpretable meaning
- › There is a **true** zero point
- › Appropriate operations: comparison, ordering, **+**, **-**, **×** and **÷**

Quantitative Variables – Data Summary Checklist

- › Let us now focus on quantitative data
- › An appropriate summary should answer basic questions, such as
 - › Range, location, scale?
 - › Any extreme values?
 - › Uni- or multimodal?
 - › Symmetric or asymmetric?
 - › Bi- or multivariate case: relationships between variables?
- › Summaries can be **graphical** or **numerical**
 - › In practice, it is often useful to employ both

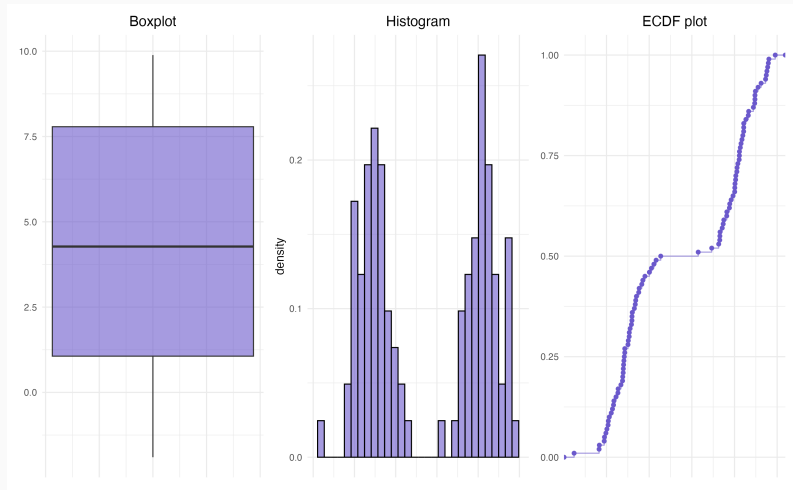
Univariate Data – Numerical Summaries

sample size		n
location	<i>mean</i>	$\bar{x} = n^{-1} \sum_{i=1}^n x_i$
	<i>α-trimmed mean</i>	$\frac{1}{n-2[\alpha n]} \sum_{j=[n\alpha]+1}^{n-[\alpha n]} x_{(j)}, \quad 0 \leq \alpha < \frac{1}{2}$
	<i>median</i>	$\text{med}(x) = \begin{cases} x_{((n+1)/2)}, & \text{if } n \text{ odd} \\ \frac{1}{2}(x_{(n/2)} + x_{(n/2+1)}), & \text{if } n \text{ even} \end{cases}$
scale	<i>variance</i>	$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$
	<i>standard deviation</i>	$s = \sqrt{s^2}$
	<i>coefficient of variation</i>	$cv = s/\bar{x}$
	<i>median absolute deviation</i>	$\frac{1}{\Phi^{-1}(3/4)} \text{med}(x_i - \text{med}(x_1, \dots, x_n))$
	<i>range</i>	$(x_{(1)}, x_{(n)})$
	<i>quartiles</i>	$\text{quart}(x), 3\text{quart}(x)$
skewness		$3\text{quart}(x) - \text{quart}(x)$
	<i>skewness</i>	$b_1 = \frac{\sqrt{n} \sum_{j=1}^n (x_j - \bar{x})^3}{\{\sum_{j=1}^n (x_j - \bar{x})^2\}^{3/2}}$
		$b_2 = \frac{n \sum_{j=1}^n (x_j - \bar{x})^4}{\{\sum_{j=1}^n (x_j - \bar{x})^2\}^2}$
size of tails	<i>curtosis</i>	

- Would you find the mean an appropriate location measure for the distribution of the height (in cm) of 2-year old children?
- What about the distribution of income for US Americans?

- Boxplot
 - Graphical counterpart of the 5-point summary
 - Can be useful to identify outliers
- Histogram
 - Shows uni- or multimodality of data (unlike the boxplot)
 - Can be useful to decide which theoretical density best fits the data
- Empirical Distribution Function
 - Shows uni- or multimodality of data (unlike the boxplot)
 - Can be compared with a theoretical cumulative distribution function

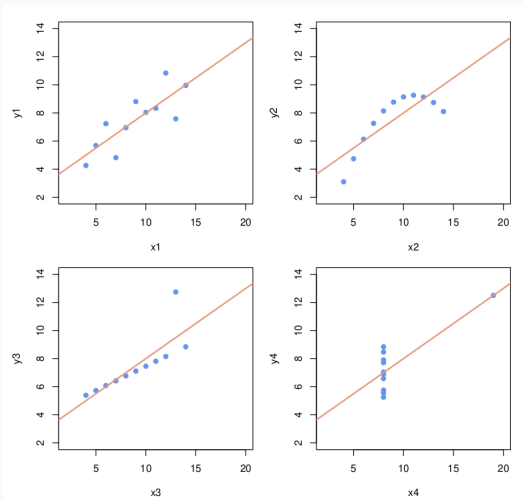
Univariate Data – Graphical Summaries Comparison



Bivariate Data – Numerical Summaries

<i>mean</i>	$(\bar{x}, \bar{y})$
<i>covariance</i>	$s_{xy} = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$
<i>correlation coefficient</i>	$r_{xy} = \frac{s_{xy}}{s_x s_y}$
<i>covariance matrix</i>	$\Sigma = \begin{pmatrix} s_x^2 & s_{xy} \\ s_{xy} & s_y^2 \end{pmatrix}$
<i>Spearman's rank correlation coefficient</i>	$r_s = \frac{\sum_{i=1}^n (r_i - \frac{1}{2}(n+1))(t_i - \frac{1}{2}(n+1))}{\sqrt{\sum_{i=1}^n (r_i - \frac{1}{2}(n+1))^2 \sum_{i=1}^n (t_i - \frac{1}{2}(n+1))^2}}$
<i>Kendall's rank correlation coefficient</i>	$\tau = \frac{\sum \sum_{i \neq j} \text{sgn}(r_i - r_j) \text{sgn}(t_i - t_j)}{n(n-1)} = \frac{4N_\tau}{n(n-1)} - 1$

Bivariate Data – Graphical Summaries



- ❗ All x vectors have the same summary statistics, same for the y vectors. The regression coefficients are also the same in all four cases. . . Plotting can sometimes be vital to get the whole picture (pun intended)!

Summarizing Multivariate Data

- Pairwise correlation coefficients
- Pairwise scatterplots

